STRAIGHT LINE

- **Distance Formula**: The distance between two points (x_1, y_1) and (x_2, y_2) is given by $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.
- **Section Formula**:

(1) **Internal Division**: If a point (x, y) divides the line joining (x_1, y_1) and (x_2, y_2) internally in the ratio $m:n$, then $x = \frac{mx_2 + nx_1}{m+n}$, $y == \frac{my_2 + ny_1}{m+n}$.

(2) **External Division**: If a point (x, y) divides the line joining (x_1, y_1) and (x_2, y_2) externally in the ratio $m:n$, then $x = \frac{mx_2 - nx_1}{m-n}$, $y == \frac{my_2 - ny_1}{m-n}$.

(3) **Mid point**: If a point (x, y) is the midpoint of the line joining (x_1, y_1) and (x_2, y_2), then $x = \frac{x_2 + x_1}{2}$, $y == \frac{y_2 + y_1}{2}$.

- **Area of triangle**: Area of a triangle with three vertices (x_1, y_1), (x_2, y_2) and (x_3, y_3) is given by $Area = \left\| \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} \right\| = \left| \frac{1}{2} [x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)] \right|$.
- **Colinearity of three points**: Three points (x_1, y_1), (x_2, y_2) and (x_3, y_3) are collinear if area bounded by them is zero. i.e. $\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$.
- **Area of a Quadrilateral**: Area of quadrilateral with four vertices (x_1, y_1), (x_2, y_2), (x_3, y_3) and (x_4, y_4) is $Area = \left| \frac{1}{2} \left[\begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix} + \begin{vmatrix} x_2 & y_2 \\ x_3 & y_3 \end{vmatrix} + \begin{vmatrix} x_3 & y_3 \\ x_4 & y_4 \end{vmatrix} + \begin{vmatrix} x_4 & y_4 \\ x_1 & y_1 \end{vmatrix} \right] \right|$.
- **Area of a Polygon**: Area of polygon with n vertices (x_1, y_1), (x_2, y_2), (x_3, y_3), ..., (x_n, y_n) is $Area = \left| \frac{1}{2} \left[\begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix} + \begin{vmatrix} x_2 & y_2 \\ x_3 & y_3 \end{vmatrix} + \begin{vmatrix} x_3 & y_3 \\ x_4 & y_4 \end{vmatrix} + \ldots + \begin{vmatrix} x_n & y_n \\ x_1 & y_1 \end{vmatrix} \right] \right|$.
- **Area of triangle bounded by three sides**: If $a_1 x + b_1 y + c_1 = 0$, $a_2 x + b_2 y + c_2 = 0$ and $a_3 x + b_3 y + c_3 = 0$ are the equation of the sides of the triangle, then area of the triangle is given by: $Area = \frac{1}{2|C_1 C_2 C_3|} \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$,

where $C_1 = a_2 b_3 - a_3 b_2$, $C_2 = a_3 b_1 - a_1 b_3$, $C_3 = a_1 b_2 - a_2 b_1$

- **<u>Locus</u>**: When a point moves in a plane under certain geometrical conditions, the point traces out a path. This path of a moving point is called its locus.

- **<u>Equation of Locus</u>**: Equation of a locus is the relation which exists between the co-ordinates of any point on the path and which holds for no other point except those lying on the path.

- **<u>Process to find the equation of the locus of a point</u>**:

(1) If we are finding the equation of the locus of a point P, assign co-ordinates $P(h,k)$.

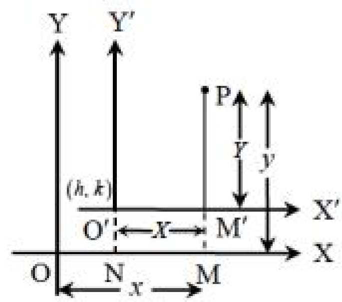
(2) Express the given conditions as equation in terms of the known quantities to facilitate calculations. We sometimes include some unknown quantities known as parameters.

(3) Eliminate the parameters, so that eliminate contains only h,k and known quantities.

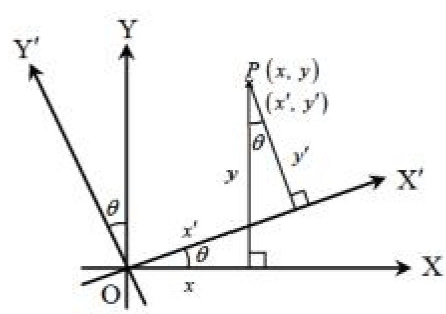
(4) Replace h by x and k by y, in the elimination. The resulting equation would be the equation of the locus of P.

(5) If x and y coordinates of the moving point are obtained in terms of third variable t. Eliminate t to obtain the relation in x & y, simplify this relation. This will give the required equation of locus.

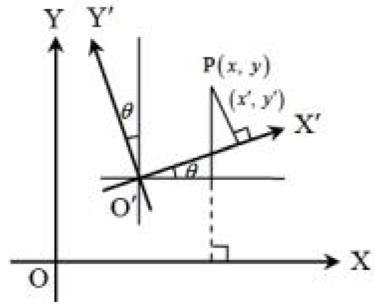
- **<u>Translation of axes</u>**: It involves shifting of origin to a new point, the new axes remaining parallel to the original axes. Let OX and OY be the original axes and new origin shifted to $O'(h,k)$. So if old coordinate was (x,y), then new coordinate is $(x-h, y-k)$.



- **<u>Rotation of Axes without changing origin</u>**: Let OX and OY be the original axes and new axes obtained OX' and OY' by rotating an angle θ in the anticlockwise direction. Let P be any point in the plane with coordinates (x,y), then new coordinate (x',y') is given by $x' = x\cos\theta + y\sin\theta,\ \ y' = -x\sin\theta + y\cos\theta$.



- **Change of origin and rotation of axes**: If origin shifted to $O'(h,k)$ and axes are rotated through an angle θ in the anticlockwise direction. Then new coordinate of (x,y) is (x',y') given by $x' = x\cos\theta + y\sin\theta - h,\ \ y' = -x\sin\theta + y\cos\theta - k$.



- **General equation of Straight line**: Equation of the form $ax + by + c = 0$, where a,b,c are constant and a,b not simultaneously zero, always represents a straight line.

- **Slope of a line**:

(1) If a line makes an angle θ with the positive direction of x axis, then $\tan\theta$ is called as gradient or slope of the line. It is denoted by $m = \tan\theta$.

(2) Slope of a line passes though two points (x_1, y_1) and (x_2, y_2) is $\dfrac{y_2 - y_1}{x_2 - x_1}$.

- **Intercepts of a line on the axes**:

(1) **Intercept of a line on x-axis**: If a line cuts x-axis at $(a,0)$, then a is called as intercept of the line on x-axis and $|a|$ is called a s length of intercept of line on x-axis.

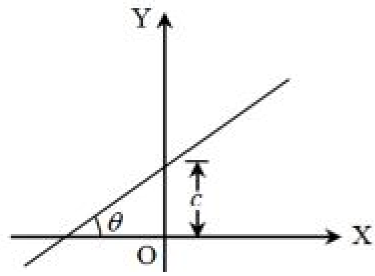
(2) **Intercept of a line on y-axis**: If a line cuts y-axis at $(0,b)$, then b is called as intercept of the line on y-axis and $|b|$ is called a s length of intercept of line on y-axis.

- **Equation of lines parallel to Axes**:

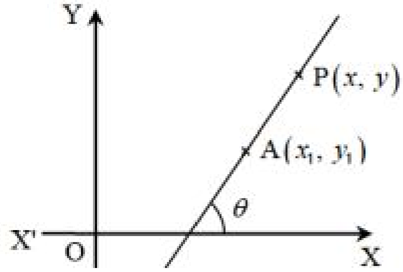
(1) Equation of x-axis is $y = 0$.
(2) Equation of y-axis is $x = 0$.
(3) Equation of a line parallel to x-axis and a unit above x-axis is $y = a$.
(4) Equation of a line parallel to x-axis and a unit below x-axis is $y = -a$.
(5) Equation of a line parallel to y-axis and a unit right of y-axis is $x = a$.
(6) Equation of a line parallel to y-axis and a unit left of y-axis is $x = -a$.

- **<u>Equation of line Various Forms</u>**:

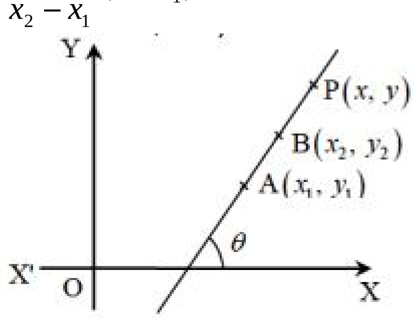
(1) **<u>Slope Intercept Form</u>**: Equation of straight line whose slope is m and y intercept is c is $y = mx + c$.



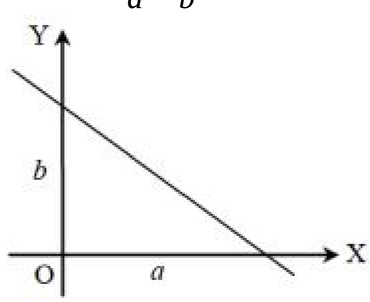
(2) **<u>Point-Slope Form</u>**: Equation of straight line whose slope is m and passes through a point (x_1, y_1) is $y - y_1 = m(x - x_1)$.



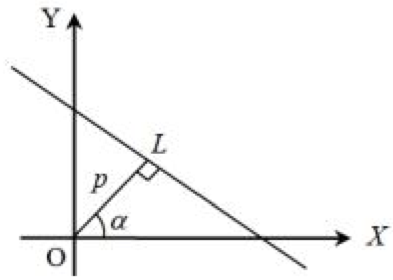
(3) **<u>Two point Form</u>**: Equation of straight line passes through two points (x_1, y_1) and (x_2, y_2) is $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$.



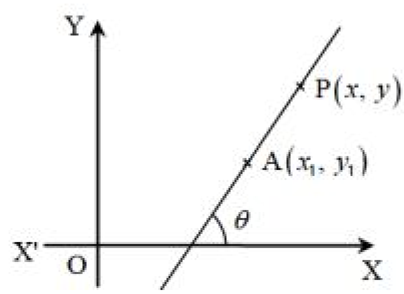
(4) **<u>Intercept Form</u>**: Equation of straight line which cuts off intercepts a and b on x-axis and y-axis respectively is $\frac{x}{a} + \frac{y}{b} = 1$.



(5) **Normal Form**: Equation of straight line upon which the length of perpendicular from the origin is p and the perpendicular makes an angle α with the positive direction of x-axis is $x\cos\alpha + y\sin\alpha = p$.



(6) **Parametric Form or Symmetric Form**: Equation of straight line passing though the point (x_1, y_1) and making an angle θ with the positive direction of x-axis is $\frac{x-x_1}{\cos\theta} = \frac{y-y_1}{\sin\theta} = r$ where $0 \le \theta \le 2\pi$ and r is the distance of the point (x, y) from the point (x_1, y_1). The co-ordinate of any point on the line at a distance r from the point (x_1, y_1) can be taken as $(x_1 + r\cos\theta, y_1 + r\sin\theta)$ or $(x_1 - r\cos\theta, y_1 - r\sin\theta)$ where $0 \le \theta \le \pi$.



- **Reduction of the general equation of straight line to different forms**: Let equation of straight line in general form is $ax + by + c = 0$.

(1) **Slope Intercept Form**:

$ax + by + c = 0 \Rightarrow y = \frac{-a}{b}x + \frac{-c}{b}$, where slope = $\frac{-a}{b}$ and y intercept = $\frac{-c}{b}$

(2) **Intercept Form**:

$$ax + by + c = 0 \Rightarrow ax + by = -c \Rightarrow \frac{a}{-c}x + \frac{b}{-c}y = 1 \Rightarrow \frac{x}{-c/a} + \frac{y}{-c/b} = 1$$

Here x intercept = $\frac{-c}{a}$ and y intercept = $\frac{-c}{b}$.

(3) **Normal Form**:

(a) **Case 1**:

When $c < 0$ or $-c > 0$, then

$$ax + by + c = 0 \Rightarrow ax + by = -c \Rightarrow \frac{a}{\sqrt{a^2+b^2}}x + \frac{b}{\sqrt{a^2+b^2}}y = \frac{-c}{\sqrt{a^2+b^2}}$$

Which is of the normal form $x\cos\alpha + y\sin\alpha = p$

Where $\cos\alpha = \frac{a}{\sqrt{a^2+b^2}}$, $\sin\alpha = \frac{b}{\sqrt{a^2+b^2}}$, $p = \frac{-c}{\sqrt{a^2+b^2}}$

(b) **Case 2**:

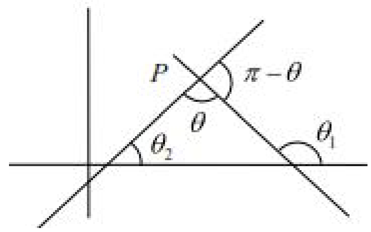
When $c > 0$ or $-c < 0$, then

$$ax+by+c=0 \Rightarrow ax+by=-c \Rightarrow \frac{a}{-\sqrt{a^2+b^2}}x+\frac{b}{-\sqrt{a^2+b^2}}y=\frac{-c}{-\sqrt{a^2+b^2}}$$

Which is of the normal form $x\cos\alpha + y\sin\alpha = p$

Where $\cos\alpha = \frac{a}{-\sqrt{a^2+b^2}}$, $\sin\alpha = \frac{b}{-\sqrt{a^2+b^2}}$, $p = \frac{c}{\sqrt{a^2+b^2}}$

- **<u>Angle between two intersecting Lines</u>**: The angle between two lines $y = m_1x + c_1$ and $y = m_2x + c_2$ is given by $\tan\theta = \pm\frac{m_1 - m_2}{1+m_1m_2} \Rightarrow \tan\theta = \left|\frac{m_1-m_2}{1+m_1m_2}\right|$, provided no line perpendicular to x-axis.



(1) If both the lines are perpendicular to x-axis, then angle between them is 0^0.

(2) If any one of the line is perpendicular to x-axis, then the slope of that line is infinite.

Let $m_1 = \infty$, then $\tan\theta = \left|\frac{m_1-m_2}{1+m_1m_2}\right| = \left|\frac{1-\frac{m_2}{m_1}}{\frac{1}{m_1}+m_2}\right| = \left|\frac{1-0}{0+m_2}\right| = \left|\frac{1}{m_2}\right|$.

(3) Two lines $y = m_1x + c_1$ and $y = m_2x + c_2$ are parallel if $m_1 = m_2$.

(4) Two lines $y = m_1x + c_1$ and $y = m_2x + c_2$ are perpendicular if $m_1 \times m_2 = -1$.

- **<u>Condition for two ;one to be coincident, parallel or perpendicular or Intersecting</u>**: Two lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are:

(1) **Coincident** if: $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$.

(2) **Parallel** if: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$.

(3) **Perpendicular** if: $a_1a_2 + b_1b_2 + +c_1c_2 = 0$.

(4) **Intersecting if:** $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$.

- **<u>Equation of a line parallel to a given Line</u>**: Equation of a line parallel to a given line $ax+by+c=0$ is $ax+by+K=0$, where K is any constant.

- **<u>Equation of a line perpendicular to a given Line</u>**: Equation of a line perpendicular to a given line $ax+by+c=0$ is $ay-bx+K=0$, where K is any constant.

- **<u>Point of intersection of two given lines</u>**: Point of intersection of two given lines $a_1x+b_1y+c_1=0$ and $a_2x+b_2y+c_2=0$ is $\left(\dfrac{b_1c_2-b_2c_1}{a_1b_2-a_2b_1}, \dfrac{c_1a_2-c_2a_1}{a_1b_2-a_2b_1}\right)$.

- **<u>Interior angles of a triangle</u>**: Let slopes three sides of the triangle are in decreasing order as $m_1 > m_2 > m_3$. Then $\tan\alpha = \dfrac{m_1-m_2}{1+m_1m_2}$, $\tan\beta = \dfrac{m_2-m_3}{1+m_2m_3}$ and $\tan\gamma = \dfrac{m_3-m_1}{1+m_3m_1}$.

- **<u>Equation of Line through the intersection of two lines</u>**: Equation of line passes through the intersection of two lines $a_1x+b_1y+c_1=0$ and $a_2x+b_2y+c_2=0$ is $a_1x+b_1y+c_1+K(a_2x+b_2y+c_2)=0$.

- **<u>Condition for three lines to be concurrent</u>**: Three lines $a_1x+b_1y+c_1=0$, $a_2x+b_2y+c_2=0$ and $a_3x+b_3y+c_3=0$ are concurrent if $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}=0$.

- **<u>Position of two point relative to a line</u>**:

  (1) Two points (x_1, y_1) and (x_2, y_2) are on the same side of the line $ax+by+c=0$ if $a_1x+b_1y+c_1$ and $a_2x+b_2y+c_2$ are of same sign.

  (2) Two points (x_1, y_1) and (x_2, y_2) are on the opposite side of the line $ax+by+c=0$ if $a_1x+b_1y+c_1$ and $a_2x+b_2y+c_2$ are of opposite sign

- **<u>Length of perpendicular distance from a point on a line</u>**: Length of perpendicular distance from a point (x_1, y_1) to the line $ax+by+c=0$ is $d=\left|\dfrac{a_1x+b_1y+c_1}{\sqrt{a^2+b^2}}\right|$.

- **<u>Distance between two parallel lines</u>**: Distance between two parallel lines $ax+by+c_1=0$ and $ax+by+c_2=0$ is $d=\left|\dfrac{c_1-c_2}{\sqrt{a^2+b^2}}\right|$.

- **<u>Area of a parallelogram or a rhombus, equation of whose sides are given</u>**:

(1) Area of Parallelogram = $\dfrac{p_1 p_2}{\sin\theta}$,

(2) Area of Rhombus = $\dfrac{{p_1}^2 p_2}{\sin\theta}$ as $p_1 = p_2$ in case of rhombus.

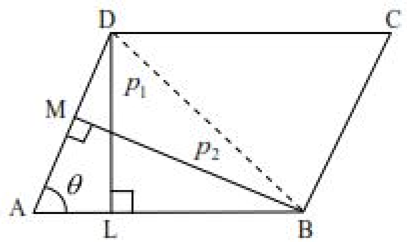
Area of Rhombus = $\dfrac{1}{2} d_1 d_2$.

$p_1 =$ DL = Distance between the lines AB and CD.

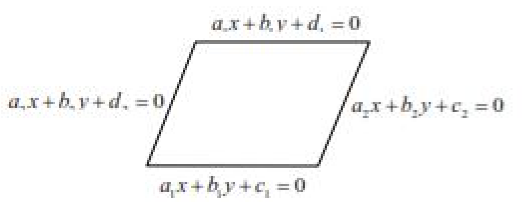
$p_2 =$ BM = Distance between the lines AD and BC.

θ = Angle between the adjacent sides AB and AD.

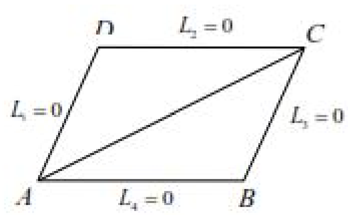
d_1, d_2 are the diagonals



(3) If the equation of sides of a parallelogram are $a_1x + b_1y + c_1 = 0$, $a_1x + b_1y + d_1 = 0$, $a_2x + b_2y + c_2 = 0$, $a_2x + b_2y + d_2 = 0$, then area is given by $\left|\dfrac{(c_1 - d_1)(c_2 - d_2)}{a_1b_2 - a_2b_1}\right|$.



(4) If the equation of sides of a parallelogram are $L_1 = 0$, $L_2 = 0$, $L_3 = 0$, $L_4 = 0$, then the diagonal BD is given by $L_2L_3 - L_1L_4 = 0$ and diagonal AC is given by $L_2L_3 - L_1L_4 = 0$.



- **<u>Equation of a straight line passes through a given point and making an angle with another given line</u>**: Equation of straight line which passes through a point (x_1, y_1) and making an angle α with a given line $y = mx + c$ are $y - y_1 = \frac{m + \tan\alpha}{1 - m\tan\alpha}(x - x_1)$ and $y - y_1 = \frac{m - \tan\alpha}{1 + m\tan\alpha}(x - x_1)$.

- **Equation of the bisector of the angle between two lines**: Equation of bisector of angle between two lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are $\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \pm \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$.
- **To find the equation of bisector of the acute and obtuse angle between two lines**: Let the equation of two lines be $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ where $c_1, c_2 > 0$.

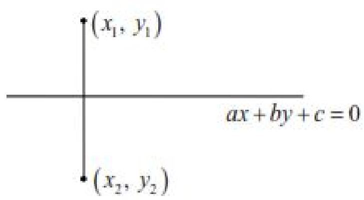
  (1) $\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = + \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$ is the equation of bisector containing origin.

  (2) $\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = - \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$ is the equation of bisector not containing origin.

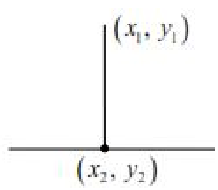
  (3) If $a_1a_2 + b_1b_2 > 0$, then the origin lie in obtuse angle and hence $\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = + \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$ is the obtuse angle bisector.

  (4) If $a_1a_2 + b_1b_2 < 0$, then the origin lie in acute angle and hence $\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = + \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$ is the acute angle bisector.

- **Image of a point**: The co-ordinate of the image (x_2, y_2) of a given point (x_1, y_1) in the line $ax + by + c = 0$ is given by $\frac{x_2 - x_1}{a} = \frac{y_2 - y_1}{b} = \frac{-2(ax_1 + by_1 + c)}{a^2 + b^2}$.

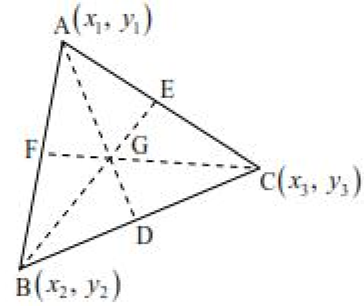


- **Foot of the perpendicular**: The co-ordinate of the foot of the perpendicular (x_2, y_2) of a given point (x_1, y_1) on the line $ax + by + c = 0$ is given by $\frac{x_2 - x_1}{a} = \frac{y_2 - y_1}{b} = \frac{-(ax_1 + by_1 + c)}{a^2 + b^2}$.



- **Standard Points of a triangle**:

(1) **Centroid of a triangle**: The point of intersection of medians of the triangle is called as centroid of the triangle.



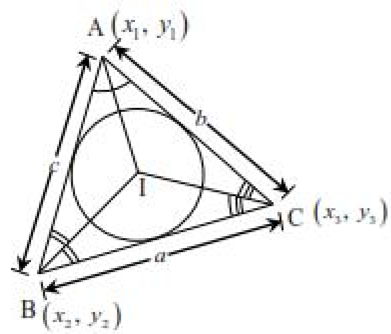
(a) Centroid divided the median in the ratio 2:1.

(b) Co-ordinate of centroid of a triangle with vertices (x_1, y_1), (x_2, y_2) and (x_3, y_3) is $\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$.

(c) If P is any point in the plane of the triangle ABC and G is the centroid then $PA^2+PB^2+PC^2 = GA^2+GB^2+GC^2+3PG^2$.

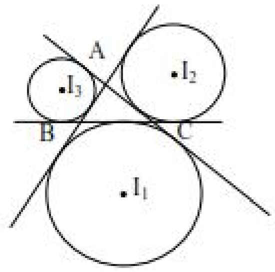
(d) If G is the centroid of the triangle ABC, then $AB^2+BC^2+CA^2 = 3\left(GA^2+GB^2+GC^2\right)$

(2) **In-centre of a triangle**: The point of intersection of angle bisector of the triangle is called as in-centre of the triangle.



(a) Co-ordinate of in-centre of a triangle with vertices (x_1, y_1), (x_2, y_2) and (x_3, y_3) is $\left(\frac{ax_1+bx_2+cx_3}{a+b+c}, \frac{ay_1+by_2+cy_3}{a+b+c}\right)$.

(3) **Ex-centre of a triangle**: A circle touches one side outside the triangle and other two extended sides, then the circle is called as Ex-circle and its centre is called as ex-centre. Let ABC is a triangle with tree ex-circles with their centres I_1, I_2, I_3 opposite to the vertices A, B and C. If vertices of the triangle ABC are $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$, then:

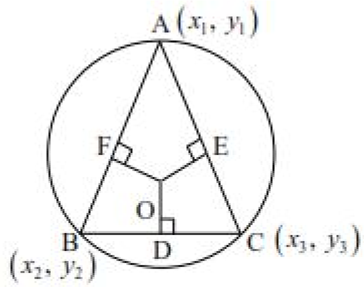


(a) $I_1 = \left(\frac{-ax_1 + bx_2 + cx_3}{-a+b+c}, \frac{-ay_1 + by_2 + cy_3}{-a+b+c} \right)$.

(b) $I_1 = \left(\frac{ax_1 - bx_2 + cx_3}{a-b+c}, \frac{ay_1 - by_2 + cy_3}{a-b+c} \right)$.

(c) $I_1 = \left(\frac{ax_1 + bx_2 - cx_3}{a+b-c}, \frac{ay_1 + by_2 - cy_3}{a+b-c} \right)$.

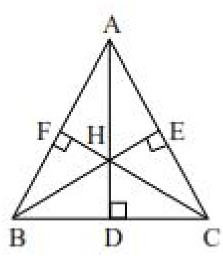
(4) **<u>Circum-centre of a triangle</u>**: The point of intersection of perpendicular bisector of sides of a triangle is called as circum-centre.



(a) If vertices of the triangle ABC are $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$, then its circum-centre is given by:

$$\left(\frac{x_1 \sin 2A + x_2 \sin 2B + x_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C}, \frac{y_1 \sin 2A + y_2 \sin 2B + y_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C} \right)$$

(5) **<u>Ortho-centre of a triangle</u>**: The point of intersection of altitudes of a triangle from the vertices to its opposite sides is called as ortho-centre.



(a) If vertices of the triangle ABC are $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$, then its ortho-centre is given by:

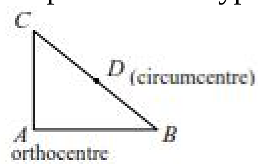
$$\left(\frac{x_1 \tan A + x_2 \tan B + x_3 \tan C}{\tan A + \tan B + \tan C}, \frac{y_1 \tan A + y_2 \tan B + y_3 \tan C}{\tan A + \tan B + \tan C} \right).$$

(b) If H is the orthocenter of triangle ABC, then C is the orthocenter of triangle HAB, B is orthocenter of triangle HAC and A is the orthocenter of triangle HBC.

(c) If any two sides out of three sides of a triangle are perpendicular, then orthocenter is the point of intersection of two perpendicular sides.

(d) In right angled triangle vertex opposite to hypotenuse is the orthocenter.

(e) In a right triangle midpoint of the hypotenuse is the circum-centre.



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